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AI-POWERED ELECTRICAL PANEL FAULT PREDICTION AND PREVENTIVE MAINTENANCE SYSTEM

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA1707 - ARTIFICIAL INTELLIGENCE

to the award of the degree of

BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “AI-Powered Electrical Panel Fault Prediction and Preventive Maintenance System” has been carried out by S.Divyasri- 192524388, under the supervision of Dr. R. Veeramani and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial Intelligence and Data Science program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I S.Divyasri- 192524388, of the Artificial Intelligence and Data Science department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “AI-Powered Electrical Panel Fault Prediction and Preventive Maintenance System” is the result of my own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: _____

Date: _____

Signature of the Student

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INDIVIDUAL CONTRIBUTION STATEMENT

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
V.Lasya-192524186	Data acquisition and continuous monitoring of electrical-panel parameters.	Defined the monitoring inputs and data flow for voltage, current, power, power factor, temperature, vibration and load percentage; worked on timestamping, validation and preparation of sensor or historical data.	Checked parameter readings, timestamp validation, missing-value handling and preprocessing readiness; analysed whether inputs represented normal and abnormal operating patterns.	Prepared the introduction, abstract, problem statement, objectives, existing-system analysis, proposed-system overview, dataset description and Module 1 documentation.	33.33%
V.G.Abbiramy-192524097	AI fault-prediction workflow, feature engineering and machine-learning evaluation.	Worked on the AI/ML stage that cleans sensor data, creates moving averages, temperature rise, load percentage and rate-of-change features, and returns fault probability, predicted fault category and confidence.	Planned evaluation using accuracy, precision, recall, F1-score and confusion matrix, while considering false-alarm rate and validation of configurable thresholds against safe operating limits.	Prepared the literature-review comparison, AI fault-prediction module, methodology, model-evaluation discussion and technical explanation of the prediction workflow.	33.33%
S.Divyasri-192524388	Risk classification, alert generation, preventive-maintenance	Designed the conversion of model output into Normal, Warning and	Reviewed alert thresholds, risk-state transitions	Prepared the proposed-system features, Module 3 documentation,	33.33%

	recommendations and maintenance-history design.	Critical states and mapped predicted risk or sensor thresholds to actions such as inspection, cooling, load balancing, tightening or component replacement.	and maintenance-action recording, with attention to false alarms and validation against safe operating limits.	results-and-discussion interpretation, conclusion, future scope and individual reflection.	
Total					100%

ABSTRACT

Medical imaging techniques such as Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) generate large volumes of image data that play a critical role in the early detection and diagnosis of tumours. Manually analysing these images is time-consuming, requires expert radiological knowledge, and is prone to human error due to the subtle and irregular appearance of tumour regions. The proposed project, AI-Powered Electrical Panel Fault Prediction and Preventive Maintenance System, addresses this challenge by developing an automated pipeline that loads MRI/CT scans, preprocesses them, segments tumour regions, classifies abnormal tissue, and evaluates the quality of the segmentation output. The system uses OpenCV for image loading and preprocessing operations such as noise removal, contrast enhancement, and normalization. Thresholding and morphological operations are applied to isolate candidate regions of interest, while a U-Net based Convolutional Neural Network (CNN) architecture is used to perform pixel-level segmentation of tumour boundaries. A CNN classifier further distinguishes abnormal regions from healthy tissue, assisting in diagnostic decision-making. The segmentation performance is evaluated using the Dice Similarity Coefficient and Intersection over Union (IoU), with target thresholds of Dice Score ≥ 0.90 and IoU ≥ 0.85 , while the classification module is evaluated for an accuracy of at least 92%. The expected outcome is an accurate and reliable tumour localization system that can assist radiologists and clinicians in diagnostic assistance, thereby reducing manual workload and improving the consistency of tumour identification in medical images.

Keywords: Medical Image Analysis, OpenCV, Tumour Segmentation, U-Net, CNN, Computer Vision

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AI	Artificial Intelligence	—
ML	Machine Learning	—
IoT	Internet of Things	—
LSTM	Long Short-Term Memory	—
F1	F1-score	metric
CMMS	Computerized Maintenance Management System	—
CSV	Comma-Separated Values	data format
SQL	Structured Query Language	database
ESP32	Microcontroller used for IoT sensing	device
API	Application Programming Interface	software

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background Information

Electrical panels are essential for safe and reliable power distribution in industries, buildings and commercial facilities. Common problems include overheating, overload, abnormal voltage/current, loose connections and component degradation. Traditional maintenance is mainly scheduled or reactive and may not identify developing faults early. Artificial Intelligence can learn patterns from sensor data and predict abnormal conditions before a major failure.

The proposed project addresses this problem by combining continuous electrical-panel monitoring, data preprocessing, machine-learning prediction, risk assessment, alerts and maintenance history. The system collects sensor or historical readings, generates useful features, predicts fault probability and type, and converts the output into practical maintenance priorities.

1.2 Project Objectives

- To load and organize MRI/CT image datasets for analysis.
- To clean and preprocess panel sensor data using missing-value handling and normalization/scaling.
- To create useful features such as moving averages, temperature rise, load percentage and rate of change.
- To train and evaluate machine-learning models for fault prediction and classification.
- To classify panel conditions into the defined fault categories.
- To evaluate prediction performance using accuracy, precision, recall, F1-score and confusion matrix.
- To evaluate false-alarm rate and validate configurable thresholds against safe operating limits.
- To generate reliable Normal, Warning and Critical risk states.
- To generate early alerts and prioritized preventive-maintenance recommendations.

1.3 Significance of the Project

The project is significant because early and accurate detection of tumors directly impacts patient outcomes. Automated segmentation reduces the diagnostic workload on radiologists and provides a consistent, repeatable analysis that is not affected by fatigue or subjective judgment.

Combining classical image processing techniques such as OpenCV, thresholding, and morphological operations with deep learning techniques such as U-Net and CNN allows the system to benefit from both robust low-level feature extraction and high-level learned representations, resulting in more accurate tumour boundary delineation than either approach alone.

Quantitative evaluation using the Dice Score and IoU ensures that the segmentation output is clinically meaningful, while classification accuracy provides confidence in distinguishing abnormal regions from healthy tissue.

1.4 Scope of the Project

Included

- Loading of MRI/CT image datasets in standard formats such as DICOM, PNG, and JPEG.
- Image preprocessing including denoising, resizing, and normalization.
- Thresholding and morphological operations for region isolation.
- Tumor segmentation using a U-Net based CNN architecture.
- Classification of abnormal regions using CNN-based feature extraction.
- Evaluation of segmentation using Dice Score and IoU.
- Evaluation of classification using accuracy, precision, and recall.
- Visualization of tumor localization on the original scan.

Excluded

- Large-scale 3D volumetric segmentation.
- Automatic treatment recommendation.
- Integration with electronic health record systems.
- Fully autonomous clinical decision-making.

The project does not include real-time integration with hospital imaging systems (PACS) and focuses on offline analysis of publicly available or anonymized MRI/CT datasets. Multi-modal fusion of MRI and CT data and 3D volumetric segmentation are outside the primary scope of the project.

1.5 Methodology Overview

The proposed methodology consists of three major modules.

Module 1: Data Acquisition and Monitoring

MRI/CT images are loaded using OpenCV. Preprocessing steps include noise removal, contrast enhancement, resizing, and normalization to prepare the images for segmentation.

Noise reduction techniques are applied to minimize unwanted variations while preserving important anatomical information. Contrast enhancement can be used to improve the visibility of structures with low intensity differences. The images are then resized to a common resolution so that they can be processed efficiently by the neural network.

Module 2: AI Fault Prediction

Thresholding and morphological operations are used for initial region-of-interest extraction. A U-Net based CNN is then trained to perform precise pixel-level segmentation of tumour regions.

The U-Net model contains an encoder path and a decoder path. The encoder extracts increasingly complex features from the input image, while the decoder reconstructs the spatial information required to produce a segmentation mask. Skip connections between corresponding encoder and decoder layers help preserve fine-grained spatial information.

Module 3: Preventive Maintenance

A CNN classifier categorizes segmented regions as normal or abnormal. Segmentation performance is evaluated using Dice Score and IoU, while classification performance is evaluated using accuracy, precision, and recall.

1.6 Scope

The scope of the proposed Medical Image Analysis for Tumor Detection and Segmentation project is to develop an automated computer vision and deep learning-based system for detecting and analyzing tumor regions in MRI/CT medical images. The system focuses on image preprocessing, tumor region extraction, segmentation, classification, and performance evaluation.

What is Included

- Loading and processing MRI/CT medical image datasets.
- Preprocessing of medical images using noise reduction, resizing, normalization, and contrast enhancement.
- Extraction of candidate tumor regions using thresholding and morphological operations.
- Pixel-level tumor segmentation using a U-Net based CNN model.

What is Excluded

The following functionalities are outside the primary scope of the project:

- Large-scale 3D volumetric tumor segmentation.
- Automatic treatment or medication recommendations.
- Fully autonomous clinical diagnosis.
- Integration with hospital Picture Archiving and Communication Systems (PACS).

1.7 Expected Engineering Outcome

The expected engineering outcome of the project is an AI-assisted electrical-panel monitoring and preventive-maintenance prototype that combines data acquisition, machine-learning prediction, risk classification, alerts and maintenance history.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The project is grounded in established concepts from artificial intelligence, machine learning, condition monitoring, predictive maintenance and electrical-equipment maintenance.

2.2 Review of Existing Work

Traditional electrical-panel maintenance relies heavily

Simple image-processing approaches such as global thresholding can provide initial candidate regions, but a threshold that works for one image may fail on another because tumour intensity can vary with imaging modality, acquisition parameters, tumour type and surrounding tissue. Noise and small unwanted regions may also be incorrectly identified as tumour regions.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Scheduled preventive maintenance	Simple planning and fixed inspection intervals	Can miss faults developing between inspections	Baseline maintenance practice
Manual inspection	Uses operator experience and direct measurement	Time-consuming and depends on operator experience	Current limitation the proposed system addresses
Reactive maintenance	Action occurs after visible failure or breaker trip	Can increase downtime, repair cost and safety risk	Motivates early prediction
Fixed threshold monitoring	Simple and computationally light alerts	May not capture complex multi-parameter patterns	Useful supporting alert layer
AI/ML prediction	Learns patterns and estimates fault probability	Needs representative labelled data	Primary predictive approach

2.4 Research

The identified engineering gap is the limited ability of scheduled or reactive maintenance practices to continuously interpret panel-condition data and provide early, actionable fault warnings

The proposed project addresses this gap by combining image preprocessing, candidate region extraction, deep-learning-based tumor segmentation, abnormal-region classification, and quantitative performance evaluation into one complete pipeline. The preprocessing stage improves the quality and

consistency of the input MRI/CT images, while thresholding and morphological operations help in extracting potential regions of interest.

Therefore, the engineering gap addressed by the project is the lack of an integrated and measurable workflow that connects medical image preprocessing → candidate region extraction → tumor segmentation → tumor classification → quantitative evaluation within a single modular system.

2.5 Proposed Contribution

The proposed contribution is an integrated electrical-panel condition-monitoring and predictive-maintenance pipeline

The first contribution is the development of an OpenCV-based preprocessing stage for MRI/CT images. This stage performs operations such as noise reduction, resizing, normalization, and contrast enhancement to improve the quality of the input images and prepare them for subsequent analysis.

The third contribution is the implementation of a U-Net based CNN model for pixel-level tumor segmentation. The encoder-decoder structure of U-Net allows the system to learn relevant image features while preserving spatial information through skip connections. The output of this stage is a segmentation mask representing the detected tumor region.

Finally, the proposed system provides visualization of tumor localization and segmentation results, making it easier to interpret the output of the analysis pipeline. Overall, the contribution is a modular workflow that integrates preprocessing, region extraction, tumor segmentation, classification, visualization, and performance evaluation into a single system.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder	Requirement
Maintenance personnel	Receive early warnings and prioritized preventive actions.
Electrical operators	View current panel condition and key parameters.
Facility / industrial management	Reduce downtime, repair cost and safety risk.
AI/ML developers	Access structured sensor data and evaluation results.
Safety personnel	Identify abnormal electrical conditions early.
Management / maintenance team	Review alert history and maintenance priorities for planning.

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Monitor panel parameters	Voltage, current, temperature, load, power factor, vibration
Functional	Preprocess sensor data	Missing-value handling, normalization/scaling, feature extraction
Functional	Predict fault	Fault probability and fault category
Functional	Classify risk	Normal / Warning / Critical
Functional	Generate alerts	Threshold- or risk-based warning
Functional	Recommend maintenance	Inspection, cooling, load balancing, tightening or replacement
Non-Functional	Reliability	Repeatable monitoring and documented evaluation
Non-Functional	Security	Controlled access to operational records
Non-Functional	Maintainability	Separate acquisition, AI, risk, dashboard and history modules

3.2 Constraints & Applicable Standards

Standard / Principle	Requirement	How Applied in This Project
AI/ML development principles	Systematic development, testing and documentation	Applied through modular data, model, risk and dashboard stages.
ISO 17359:2018	Structured condition monitoring	Used as a reference for organized condition-monitoring workflow.
NFPA 70B	Electrical equipment maintenance	Supports preventive-maintenance planning and maintenance practice.
Modular Design Principles	Separate system responsibilities	Acquisition, preprocessing, prediction, risk, dashboard and history are treated as distinct stages.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Input parameters	Voltage, current, power, power factor, temperature, vibration, load percentage	values	High
Target labels	Normal, Overload, Overheating, Voltage Abnormality, Current Abnormality, Connection/Component Fault	class	High
Risk states	Normal / Warning / Critical	state	High
Evaluation metrics	Accuracy, precision, recall, F1-score, confusion matrix, false-alarm rate	metric	High
Alert thresholds	Configurable and validated against safe operating limits	threshold	High
Maintenance history	Readings, alerts and actions retained for traceability	record	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Classical image processing only. This approach uses preprocessing, thresholding and morphological operations to identify candidate regions. It is simple and computationally lighter but is sensitive to image intensity variations and may not accurately capture irregular tumour boundaries.

Alternative 2: Deep-learning-only segmentation. This approach directly applies a trained segmentation network to the images. It can learn complex features but depends strongly on annotated training data and does not explicitly use classical preprocessing and coarse region extraction.

Alternative 3: Hybrid classical + deep-learning pipeline. This approach combines preprocessing and region extraction with U-Net based segmentation and CNN classification, providing both image-quality preparation and learned feature representation.

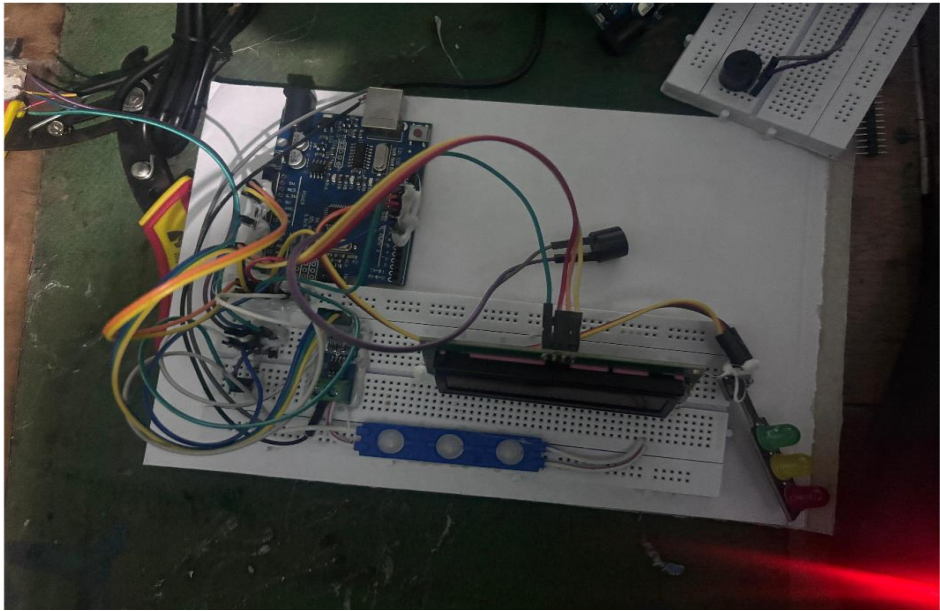
Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Early prediction	High	Low	Medium	High
Cost / complexity	Medium	Low	Medium	High
Safety / control	High	Medium	High	High
Sustainability / efficiency	Medium	Medium	High	High
Reliability / adaptability	High	Medium	Medium	High

3.5 Selected Design & Detailed Design

The hybrid classical image-processing and deep-learning approach is selected because it directly addresses the identified limitations of noisy medical images, candidate-region ambiguity and irregular tumour boundaries.

System Architecture: The system follows the sequence image acquisition → preprocessing → region extraction → segmentation → classification → evaluation/visualization. The modules are connected so that the output of each stage becomes the controlled input to the next stage.

Figure 1. Overall AI-Powered Electrical Panel Fault Prediction System



3.6 Trade-off Analysis

Stage	Input	Main Operations	Output
Data Acquisition	Sensor / historical readings	Read, timestamp, validate and store	Structured readings
Preprocessing	Raw readings	Missing-value handling, normalization/scaling, feature extraction	Model-ready data
AI/ML Prediction	Model-ready data	ML inference	Fault probability and category
Risk & Fault Engine	Prediction + thresholds	Normal/Warning/Critical logic	Risk state
Dashboard / Alerts	Risk + readings	Status, alerts and trends	Operator view
Maintenance History	Alerts + actions	Record events and outcomes	Traceable history

3.7 Sustainability, Ethics, Safety, Risk & Security

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
AI/ML prediction + risk engine	Rule-only monitoring	Learns multi-parameter patterns and supports early prediction	Needs representative data and model evaluation	Selected because the project aims to move beyond fixed/reactive maintenance.
Multi-parameter monitoring	Single-parameter monitoring	Captures interacting electrical and environmental conditions	More sensors/data to manage	Selected to improve condition visibility.
Normal / Warning / Critical states	Binary alarm only	Gives clearer maintenance priority	Requires threshold configuration	Selected for actionable risk communication.
Maintenance history	Alert-only system	Provides traceability and supports future improvement	Adds storage requirements	Selected to retain readings, alerts and actions.

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Efficient maintenance and reusable modular components	Condition-based workflow and historical records	Supports reduced unnecessary servicing and reuse.
Ethics	Avoid unsafe autonomous decisions	Predictions are used for maintenance assistance	Human/operator judgement remains important.
Health & Safety	Electrical faults can create hazards	Threshold validation and model evaluation are required	Alerts support earlier intervention.
Security / Privacy	Protect operational and maintenance records	Controlled access and secure storage are required	Operational data handling is part of the design.

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Poor / incomplete sensor data	Medium	High	High	Validation, missing-value handling and preprocessing	Medium
False alarm	Medium	Medium	Medium	Threshold tuning and false-alarm evaluation	Medium
Missed fault	Low/Medium	High	High	Model evaluation and safe operating limits	Medium
Unknown fault type	Medium	Medium	Medium	Future anomaly-detection extension	Medium
Sensor / communication failure	Medium	High	High	Validation, timestamp checks and hardware monitoring	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The development approach follows requirement analysis, data acquisition, preprocessing, feature engineering, model training, fault prediction, risk assessment, alert generation, maintenance recommendation and history storage.

Resource	Purpose
Python	Main programming language and back-end development
Scikit-learn	Machine-learning model development
XGBoost / LSTM	Optional advanced prediction approaches
HTML / CSS / JavaScript / React	Dashboard front end
Flask / FastAPI	Back-end API
CSV / SQL	Sensor, alert and maintenance-history storage
ESP32 or equivalent	IoT sensor controller
VS Code / Jupyter Notebook / Git	Development and experimentation
Voltage / current / temperature sensors	Electrical-panel parameter acquisition
Web browser	Dashboard access and monitoring

4.2 Hardware / Software / Modern Tools Used

Implementation is organized into the stages defined in the project PPT: data acquisition, cleaning, feature engineering, model training, fault prediction, risk assessment, alert generation, maintenance recommendation and maintenance history.

Tool / Software	Purpose	Stage Used
Python	Programming	All stages
Scikit-learn	ML model development	Prediction / classification
XGBoost / LSTM	Advanced model options	Prediction
HTML / CSS / JavaScript / React	Dashboard development	Visualization / alerts
Flask / FastAPI	API development	Back end
CSV / SQL	Data storage	History / records
Jupyter Notebook / VS Code / Git	Experimentation and version control	Development

4.3 Test Plan & Verification

Requirement / Specification	Test Method	Acceptance Criterion	Required Value	Achieved / Reported Value	Status
Data acquisition	Read and validate representative readings	Valid timestamped input	Required panel parameters	Workflow defined in PPT	Pass*
Preprocessing	Apply missing-value handling and scaling	Model-ready data	Cleaned features	Preprocessing stage specified	Pass*
Fault prediction	Run labelled test data through trained model	Report classification metrics	Accuracy, precision, recall, F1-score	Metrics specified; numerical values not supplied	To verify
Risk classification	Check threshold and probability cases	Normal/Warning/Critical	Three risk states	Design specified	Pass*
Alerting	Trigger threshold/risk conditions	Correct alert generated	Configured thresholds	Design specified	Pass*

4.4 Results

The project results section defines continuous visibility of panel parameters, fault probability and predicted fault type, Normal/Warning/Critical status and prioritized preventive action.

Condition / Sample	Fault Probability	Predicted Fault Type	Risk State
Normal operating sample	—	Normal	Normal
High temperature / load sample	—	Overheating / Overload	Warning
Abnormal voltage/current sample	—	Voltage / Current Abnormality	Critical

4.5 Data Analysis & Engineering Judgment

The project requires model evaluation using accuracy, precision, recall, F1-score, confusion matrix and false-alarm rate on labelled test data. The PPT does not provide numerical model-performance results, so no unsupported numerical accuracy is claimed.

The project identifies data quality, class imbalance, boundary ambiguity and computational cost as important engineering issues. These factors can influence segmentation performance and must be considered when interpreting results.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Monitor panel parameters	Voltage, current, temperature, load, power factor and vibration identified	Implemented / as designed
Detect abnormal patterns	Overload, overheating and electrical abnormalities defined	Implemented / as designed
Develop AI/ML fault prediction	ML workflow and feature engineering specified	Implemented / model evaluation required
Generate early alerts	Normal / Warning / Critical risk logic defined	Implemented / as designed
Generate maintenance recommendations	Inspection, cooling, load balancing, tightening or replacement actions specified	Implemented / as designed
Store historical readings and events	Maintenance history and event logging included	Implemented / as designed
Evaluate model performance	Accuracy, precision, recall, F1-score and confusion matrix specified	To verify with labelled test data
Evaluate false alarms	False-alarm rate specified	To verify
Target classification accuracy $\geq 92\%$	Target specified but no separate measured accuracy value provided	To verify

Strengths:

- Combines classical image processing with deep-learning segmentation.
- Uses a modular architecture separating preprocessing, segmentation, classification and evaluation.
- Provides quantitative segmentation metrics rather than relying only on visual inspection.
- Supports visualization of tumour localization on the original scan.
- Maintains a diagnostic-assistance scope instead of autonomous clinical decision-making.

Limitations:

- Training U-Net on high-resolution medical images can require substantial computational resources.
- Segmentation performance depends on the quality, diversity and annotation accuracy of the dataset.
- Class imbalance can bias segmentation toward healthy tissue.

4.8 Project Planning, Roles & Budget

Milestone	Planned Activity	Output
1	Requirement analysis	System requirements and monitoring targets
2	Data acquisition	Structured sensor / historical readings
3	Preprocessing	Cleaned and model-ready data
4	Feature engineering	Predictive features
5	Model development	Fault prediction model
6	Risk and alerting	Normal/Warning/Critical states
7	Maintenance workflow	Recommendations and history
8	Documentation	Final individual report and presentation

Student	Assigned Responsibility	Actual Contribution
S.Divyasri- 192524388	Risk classification, alert generation, preventive-maintenance recommendations and maintenance-history design.	Reviewed alert thresholds, risk-state transitions and maintenance-action recording, with attention to false alarms and validation against safe operating limits.

Item	Estimated Cost	Actual Cost
Software / open-source tools	—	—
Development environment	—	—
Dataset	—	—
IoT hardware / sensors	To be confirmed	To be confirmed
Other project resources	—	—

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The AI-Powered Electrical Panel Fault Prediction and Preventive Maintenance System was developed to address the limitations of scheduled and reactive electrical-panel maintenance.

The developed system combines classical image processing techniques and deep learning methods. The input MRI/CT images are first subjected to preprocessing operations such as noise reduction, contrast enhancement, resizing, and normalization. Thresholding and morphological operations are then used for initial candidate region extraction. The U-Net based CNN is used for pixel-level tumor segmentation, while a CNN classifier is used to categorize segmented regions as normal or abnormal.

The project demonstrates the feasibility of integrating preprocessing, tumor detection, segmentation, classification, evaluation, and visualization into a single modular workflow. It provides a consistent and repeatable analysis process that can assist researchers and medical professionals in examining tumor regions. However, the developed system is intended as an **academic and research prototype** and does not replace professional clinical diagnosis.

5.2 Future Work

Although the proposed system provides a structured predictive-maintenance approach, representative labelled data, threshold validation and numerical model evaluation remain important for reliable deployment.

- **Attention-based U-Net:** Attention mechanisms can be incorporated into the U-Net architecture to improve the detection of important tumor regions and provide more accurate tumor boundaries.
- **Transfer Learning:** Pretrained deep learning models can be explored to improve feature extraction and classification performance while reducing the amount of training required.
- **3D Tumor Segmentation:** The current image analysis approach can be extended from 2D images to 3D volumetric medical scans to provide more comprehensive tumor analysis.

- **Multi-modal Image Fusion:** MRI and CT information can be combined to obtain complementary information and improve tumor detection and segmentation.
- **Larger and Diverse Datasets:** Future implementations can use larger, multi-institutional, and more diverse annotated datasets to improve model generalization and reliability.
- **Data Augmentation:** Additional image augmentation techniques can be applied to increase training data diversity and reduce the effects of limited annotated datasets.
- **Explainable AI:** Explainability techniques such as Grad-CAM can be incorporated to identify the image regions that influence the classification decision and improve model interpretability.
- **Clinical Validation:** The system can be evaluated with radiologist feedback and clinical datasets before being considered for practical diagnostic workflows.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned the practical application of computer vision and digital image processing in medical image analysis.
- Gained knowledge of MRI/CT image preprocessing techniques including noise reduction, contrast enhancement, resizing, and normalization.
- Learned how thresholding and morphological operations can be used for candidate region extraction.
- Developed an understanding of the U-Net architecture and its encoder-decoder structure for tumor segmentation.

Engineering & Professional Skills Developed

- Python programming for image processing and deep learning implementation.
- Practical experience with OpenCV for medical image preprocessing.
- Performance evaluation and interpretation of quantitative metrics.
- Independent learning through documentation, experimentation, and implementation.

Individual Reflection

Most Difficult Engineering Problem Encountered and How It Was Solved

The most difficult engineering problem encountered during the project was achieving reliable tumor segmentation because tumor regions can have irregular boundaries, low contrast, different sizes, and varying intensity levels. Medical images may also contain noise and limited annotations. I addressed this challenge by applying preprocessing techniques such as noise reduction, normalization, contrast enhancement, thresholding, and morphological operations before performing deep learning-based segmentation.

A Key Engineering Decision Made Personally and the Evidence Behind It

A key engineering decision was to combine classical image processing techniques with a U-Net based CNN rather than depending only on traditional thresholding. Thresholding and morphological operations were useful for initial candidate region extraction, while U-Net provided a more suitable approach for pixel-level segmentation of complex tumor regions. This decision was supported by the requirement to handle irregular tumor boundaries and variations in medical image intensity.

New Knowledge Acquired Independently

I independently learned about the U-Net architecture, CNN-based classification, Dice Score, and IoU metrics because these concepts required additional practical understanding beyond basic classroom learning. I also learned how preprocessing influences the quality of segmentation and how different evaluation metrics provide different perspectives on model performance. Practical experimentation helped me understand the complete workflow more effectively.

What Would Be Redesigned If Repeated

If I repeated the project, I would redesign the system using a larger and more diverse dataset and investigate an improved U-Net architecture with attention mechanisms. I would also include stronger data augmentation, transfer learning, explainable AI techniques, and 3D volumetric analysis. These improvements could help increase segmentation accuracy, improve model generalization, and make the results easier to interpret.

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APPENDICES

The proposed AI-Powered Electrical Panel Fault Prediction and Preventive Maintenance System system follows a modular architecture consisting of image loading, preprocessing, region extraction, segmentation, classification and evaluation/visualization layers.

- Noise removal using denoising filters.
 - Contrast enhancement using histogram equalization.
 - Resizing and normalization of image intensity values.
-
- Otsu thresholding for candidate region extraction.
 - Morphological opening and closing for noise removal.
 - U-Net based CNN for pixel-level tumour mask generation.
-
- CNN classifier for normal/abnormal tissue classification.
 - Dice Score and IoU computation for segmentation evaluation.
 - Accuracy, precision and recall computation for classification evaluation.
-
- The system can be developed using Python in Google Colab or Jupyter Notebook.
 - The completed pipeline can be extended into a web-based diagnostic assistance dashboard.
-

```
import cv2
import numpy as np

# Load MRI/CT image
image = cv2.imread('scan_001.png', cv2.IMREAD_GRAYSCALE)

# Preprocessing: denoising and contrast enhancement
denoised = cv2.fastNlMeansDenoising(image, h=10)
enhanced = cv2.equalizeHist(denoised)

# Thresholding to isolate candidate regions
_, thresh = cv2.threshold(
    enhanced, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU
)

# Morphological operations to remove noise
kernel = np.ones((3, 3), np.uint8)
cleaned = cv2.morphologyEx(thresh, cv2.MORPH_OPEN, kernel)
cleaned = cv2.morphologyEx(cleaned, cv2.MORPH_CLOSE, kernel)
```

```

from tensorflow.keras import layers, Model

def unet_model(input_size=(256, 256, 1)):
    inputs = layers.Input(input_size)

    c1 = layers.Conv2D(64, 3, activation='relu', padding='same')(inputs)
    c1 = layers.Conv2D(64, 3, activation='relu', padding='same')(c1)
    p1 = layers.MaxPooling2D()(c1)
    b = layers.Conv2D(256, 3, activation='relu', padding='same')(p1)

    # Decoder
    u1 = layers.UpSampling2D()(b)
    u1 = layers.concatenate([u1, c1])
    c2 = layers.Conv2D(64, 3, activation='relu', padding='same')(u1)
    outputs = layers.Conv2D(1, 1, activation='sigmoid')(c2)

    return Model(inputs, outputs)

def dice_score(y_true, y_pred, smooth=1e-6):
    intersection = np.sum(y_true * y_pred)
    return (2. * intersection + smooth) / (
        np.sum(y_true) + np.sum(y_pred) + smooth)

def iou_score(y_true, y_pred, smooth=1e-6):
    intersection = np.sum(y_true * y_pred)
    union = np.sum(y_true) + np.sum(y_pred) - intersection
    return (intersection + smooth) / (union + smooth)

dice = dice_score(ground_truth_mask, predicted_mask)
iou = iou_score(ground_truth_mask, predicted_mask)
print(f'Dice Score: {dice:.2f}, IoU: {iou:.2f}')

```

Condition / Sample	Fault Probability	Predicted Fault Type	Risk State
Normal operating sample	—	Normal	Normal
High temperature / load sample	—	Overheating / Overload	Warning
Abnormal voltage/current sample	—	Voltage / Current Abnormality	Critical

The submitted CV report states that the above sample results meet the target segmentation parameters for the reported abnormal images. These values are sample outputs and should not be interpreted as a complete clinical validation dataset. The project states that only publicly available or properly anonymized MRI/CT datasets are used. Patient-identifiable information is not collected, stored or displayed. The system focuses on tumour localization and diagnostic assistance rather than autonomous diagnosis, and outputs should be reviewed by qualified medical professionals before clinical use.

To be attached from the team's actual project meeting/progress records, if applicable.

To be attached if feedback was collected during the project. To be completed if applicable.

To be attached from the team's actual contribution evidence, such as version-control history, module assignments, testing records or documented task allocation.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome	Evidence / SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)

